Year 9 Science – Booklet 10: Chemistry

Chemical Changes (Explaining Chemical Changes) — ANSWERS

Quick Test (p.59)

1. It is a (weak) acid – pH 5 is below 7.
2. Neutralisation is the reaction between an acid and a base, in which they react until a salt and water are formed and no acid or base remains, leaving a neutral solution (pH 7).
3. acid + metal → salt + hydrogen
4. They measure the pH directly and give a precise numerical reading, rather than relying on comparing a colour by eye, so they are more accurate than indicator paper or solution.

Vocabulary Builder (9 marks)

1. Match key word to definition

Neutral → a substance that makes a solution with a pH = 7
Alkali → a substance that reacts with an acid, that can dissolve in water to make a solution with a pH > 7
Acid → a substance that contains hydrogen and can dissolve in water to make a solution with a pH < 7
Salt → a substance made in a neutralisation reaction
Base → a substance that reacts with an acid

2. Complete the sentences about combustion (5 marks)

Combustion reactions involve **burning** a fuel with oxygen. The stored chemical **energy** in the fuel is transferred into heat and light energy. **Hydrocarbons** are fuels that contain only hydrogen and carbon. Many everyday fuels like **petrol** in cars and **natural gas** used to heat our homes are hydrocarbon fuels.

3. What is an indicator? (2 marks)

A substance that changes colour depending on whether it's in acidic, neutral or alkaline conditions, used to find out what type of substance (or the pH) a chemical is.

4. Classify the reaction type

a). hydrocarbon + oxygen → carbon dioxide + water – **Complete combustion**

b). acid + alkali → salt + water – **Neutralisation**

c). calcium + oxygen → calcium oxide – **Oxidation**

d). metal carbonate + acid → salt + carbon dioxide + water – **Neutralisation**

e). hydrocarbon + oxygen → carbon dioxide + carbon + carbon monoxide + water – **Incomplete combustion**

5. Medicine for indigestion/heartburn

Tick **Antacid**.

6. True or False about combustion

a) You only need fuel and oxygen for combustion – **False** (you also need a source of heat/ignition – the fire triangle is fuel, oxygen AND heat)
b) Combustion is exothermic – **True**
c) Hydrogen gas can fuel rockets – **True**
d) Carbon monoxide is a greenhouse gas – **False** (it's toxic, not a greenhouse gas – carbon dioxide is the greenhouse gas)
e) Soot comes from incomplete combustion of hydrocarbons – **True**

Maths Skills (11 marks)

1. Sunil's marble chips + HCl mass-vs-time graph

a). Starting mass (mass at t = 0) = **163.8 g**.

b). Mass of CO■ released = starting mass – final (plateau) mass = 163.8 – 161.4 = **2.4 g**.

c). The graph runs from 0 to 20 minutes: 20 × 60 = **1200 seconds**.

d). 0.01 kg = **10 g** (given directly in the question).

2. pH scale (1 unit = ×10 difference in acidity)

a). pH 2 vs pH 3: **10 times** more acidic.

b). pH 2 vs pH 4: **100 times** more acidic.

c). pH 2 vs pH 5: **1000 times** more acidic.

d). pH 2 vs pH 6: **10 000 times** more acidic.

3. Nitric acid production (28 million tonnes/year, pie chart)

a). **28 000 000 tonnes.**

b). Manufacture of fertilisers (the largest slice of the pie chart).

c). 10% of 28 000 000 = 2 800 000 tonnes = **2.8 × 10■ tonnes**.

d). Nylon manufacture = 1/16 of the pie chart = 1 ÷ 16 = **6.25%**.

Testing Understanding (40 marks)

1. Table of acids and their salts (4 marks)

Name of acid	Name of salt	Example
Hydrochloric acid	c) Chloride	Sodium chloride
a) Sulfuric acid	Sulfate	Copper sulfate
b) Nitric acid	d) Nitrate	Magnesium nitrate

2. Complete the word equations (6 marks)

a). calcium + **oxygen** → calcium oxide

b). ethanol + oxygen → **water** + carbon dioxide

c). **Magnesium** + hydrochloric acid → magnesium chloride + hydrogen

d). sodium carbonate + **hydrochloric acid** → carbon dioxide + water + sodium chloride

e). potassium hydroxide + sulfuric acid → **potassium sulfate** + **water**

3. Fire triangle and combustion of methane

a). A: **Oxygen**. B: **Heat**. (Fire triangle = fuel, oxygen, heat.)

b). methane + oxygen → carbon dioxide + water (complete combustion)

c). methane + oxygen → carbon monoxide + carbon + water (incomplete combustion, from insufficient oxygen)

3. d) Match pollutants to problems (3 marks)

Soot (carbon) → an irritant that can cause breathing difficulties
Carbon monoxide → a toxic gas that can lead to poisoning and death
Sulfur dioxide and nitrogen oxides → acidic gases that cause acid rain
Carbon dioxide → a greenhouse gas linked to climate change

4. a) General equations for acid + base reactions

i). metal + acid → salt + hydrogen

ii). metal carbonate + acid → salt + water + carbon dioxide iii). metal hydroxide + acid → salt + water

iv). metal oxide + acid → salt + water

4. b) Gas tests

i). Hydrogen: hold a lit splint at the mouth of the test tube – if hydrogen is present, it ignites with a squeaky pop.

ii). Carbon dioxide: bubble the gas through limewater – if carbon dioxide is present, the limewater turns cloudy/milky.

5. Sodium carbonate + hydrochloric acid experiment

a). sodium carbonate + hydrochloric acid → sodium chloride + water + carbon dioxide

b). i) Hydrochloric acid (strong acid): **Red**. ii) Sodium carbonate (alkali): **Blue/indigo-violet**. iii) Solution at the end (neutralised, pH 7): **Green**.

c). Because carbon dioxide gas was being produced by the reaction and escaping from the solution as bubbles.

d). Neutralisation (an acid reacting with a carbonate).

Working Scientifically (23 marks)

1. Jesse's marble chips + HCl mass-loss experiment

a). A measuring cylinder.

b). A conical flask.

c). The cotton wool plugs the flask's neck to stop acid spray/marble dust escaping, while still letting the carbon dioxide gas produced escape freely, so pressure doesn't build up and the balance reading stays accurate.

d). The mass of the flask and its contents (the dependent variable – what's measured).

e). The mass will decrease over time (4 marks): carbon dioxide gas is produced and escapes into the air through the cotton wool, so the total mass in the flask falls. It will fall quickly at first, while the reaction is fastest, then level off once the reaction finishes (when either the acid or the marble chips has been fully used up), after which the mass stays constant.

2. Daiyu's universal indicator results

a). Add a few drops of universal indicator (or dip UI paper) into a sample of each liquid; observe the colour it turns; compare that colour against a standard pH colour chart to read off the pH; repeat for each liquid, using a fresh/clean sample each time to avoid contamination.

b) Results table:

Liquid	pH	Colour
Lemon juice	2	Red
Cola	3	Orange
Tomato juice	4	Yellow
Water	7	Green
Baking soda	9	Blue

3. Premal's antacid investigation

a). The colour the universal indicator turns with each antacid (looking for which gets closest to green/neutral, pH 7), and/or how much of each antacid is needed to neutralise a fixed volume of acid – the best antacid needs the smallest dose to neutralise it.

b). Use the same volume and concentration of acid, the same recommended dose of each antacid, the same volume of indicator, and the same temperature for every test; use a fresh sample of acid for each antacid tested.

c). Wear eye protection and a lab coat/apron; handle the hydrochloric acid carefully, avoiding contact with skin and eyes, and wash hands afterwards.

d). Litmus only shows whether a solution is acidic or alkaline (red or blue) without giving a precise pH value across a range of colours, so it couldn't show the gradual change in acidity or precisely when neutral (pH 7) was reached, unlike universal indicator.

Science in Use (14 marks)

1. Farmers adding lime (calcium oxide) to soil

a). A base (metal oxide).

b). Neutralisation.

c). Universal indicator (solution or paper).

d). A pH probe/meter (with a data logger).

e). calcium oxide + water → calcium hydroxide

f). An alkali (a strong alkali, since pH > 7).

2. Natural gas (methane) combustion in the home

a). Methane (CH■) is a hydrocarbon because it's made only of hydrogen and carbon atoms chemically bonded together.

b). Oxygen.

c). The black powder is soot (carbon), produced when methane undergoes incomplete combustion (from a faulty boiler not getting enough oxygen), which makes carbon/carbon monoxide instead of fully burning to carbon dioxide and water. This is a serious concern because incomplete combustion also produces carbon monoxide – a toxic, colourless, odourless gas that can build up unnoticed and cause fatal poisoning; the soot is a warning sign the boiler needs urgent servicing.